

BRAC University, Dhaka
Department of Computer Science and Engineering

CSE 446: Blockchain and Cryptocurrencies, Spring 2026, Quiz - 4

Marks: 20

Time: 20 Min

Name:	ID:	Section:
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1. Determine if the following sentences are true or false. For any false sentence, write its correct form. 3*2 = 6
 - i. Currently, Ethereum uses PoW consensus algorithm. (F)
Uses PoS blockchain.
 - ii. A transaction in Ethereum changes its state. (T)
 - iii. A certificate in Fabric contains the private key of the user. (F)
Contains the public key of the user.

2. Fill in the blanks 3*2 = 6
 - i. An endorsement in Fabric represents ____ the signed R/W set created by the endorsers _____

 - ii. The node which is responsible for creating endorsements and creating blocks are _____ endorsing peers _____ and _____ ordering service node _____ respectively
 - iii. Two fields of a Fabric block header are ____ Header/current block hash/previous block hash _____

3. Briefly answer the following questions. 4*2 = 8
 - i. Write down two important requirements for private blockchain systems.
 - ii. Differentiate between a committing peer and endorsing peer.

Nodes and roles

- Committing Peer
 - Maintains the state of the Blockchain and commits transactions
 - It verifies endorsements (rules and policies) and validates the results of a transaction
- Endorsing Peer
 - An endorsing peer is always also a committing peer
 - The difference is that an endorsing peer additionally takes transaction proposals (explained later) and executes them to create endorsements (explained later)

iii. How mixHash is generated in Ethereum ?

Ethash algorithm

- Then, the latest block header and the current candidate nonce are hashed using Keccak (SHA-3) hash function
- The resultant hash is mixed several times with data from the DAG, the mixHash data field
- The final hashed digest is compared to the difficulty target and accepted or discarded accordingly

iv. Briefly discuss the attestation process in current Ethereum consensus algorithm.

Ethereum PoS algorithm

- Ethereum moved to a PoS consensus algorithm in 2022
- Like any PoS algorithm there are a number of validators
- To be a validator, one has to deposit 32 eth to an escrow contract
- On depositing their ETH, the user joins an activation queue that limits the rate of new validators joining the network
- Once activated, validators receive new blocks from peers on the Ethereum network
- The transactions delivered in the block are re-executed, and the block is verified
- The validator then sends a vote (called an attestation) in favour of that block across the network